Indian Farmer Scheme Assistant

Citation-grounded retrieval over official Indian agriculture sources

Project summary	Verified result
Corpus	406 chunks from seven official source items
Coverage	PM-KISAN, PMFBY, Soil Health Card, e-NAM, KCC
Evaluation	18 questions | Hit@1 1.000 | Citation completeness 100%
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Executive brief

Official farmer-scheme information is distributed across ministry PDFs, programme portals, FAQs, and RBI circulars. This project creates a compact evidence-first assistant that retrieves a relevant official passage and keeps the source visible.

Design choice	Reason
Official-source registry	Limits the corpus to traceable publishers
Hybrid lexical retrieval	Balances exact policy phrases and spelling variation
Extractive answers	Reduces unsupported paraphrasing of financial rules
Page-aware citations	Keeps the supporting passage inspectable
Mandatory warning	Reminds users to verify current rules and deadlines

Project value

- Demonstrates document ingestion, retrieval, evaluation, and safety design.
- Handles English, Hindi, and Hinglish test queries within a declared scope.
- Reports a small curated test honestly rather than claiming general accuracy.

Problem, scope, and intended users

The assistant supports source discovery and passage retrieval. It does not decide eligibility, submit an application, recommend a loan or insurance product, or replace the latest official portal.

Scheme area	Indexed official material
PM-KISAN	Revised operational guidelines
PMFBY	Revised crop-insurance operational guidelines
Soil Health Card	Official FAQs on cycle, parameters, and sampling
e-NAM	Government of India market explainer
Kisan Credit Card	Reserve Bank of India master circular

Primary users

- A farmer or student locating the official source for a scheme question.
- A project reader checking how citations and safety controls are implemented.
- A student studying applied NLP and retrieval fundamentals.

Source corpus and provenance

The source manifest stores a stable ID, title, publisher, URL, and document type for every item. Acquisition records a SHA-256 checksum. Raw documents and extracted chunks remain local so the repository does not redistribute full government publications.

Control	Implementation
Publisher boundary	Ministries, programme portals, PIB, and RBI
Version trace	Source title and URL stored in a versioned manifest
Acquisition trace	Checksum recorded after download
Page lineage	PDF page number retained on every extracted chunk
Repository size	Raw documents and trained index are ignored and rebuilt

Source freshness remains a risk. Rebuilding refreshes documents, but the user must still verify publication dates, eligibility, amounts, and deadlines on the cited portal.

Ingestion, extraction, and chunking

Stage	Method	Output
Acquire	Download from manifest and hash	Local raw file and checksum
Extract PDF	Read page by page	Text with page metadata
Extract HTML	Remove scripts and presentation elements	Main content text
Normalise	Collapse whitespace and clean boilerplate	Consistent text
Chunk	About 1,100 characters with overlap	406 evidence chunks

Engineering lessons

- A dynamic e-NAM page returned navigation rather than the visible FAQ content.
- A broad HTML cleanup rule initially removed RBI content wrapped in a form.
- Per-source chunk counts exposed both ingestion problems before evaluation.
- A static PIB explainer and a targeted RBI content selector corrected the corpus.

Hybrid retrieval and answer composition

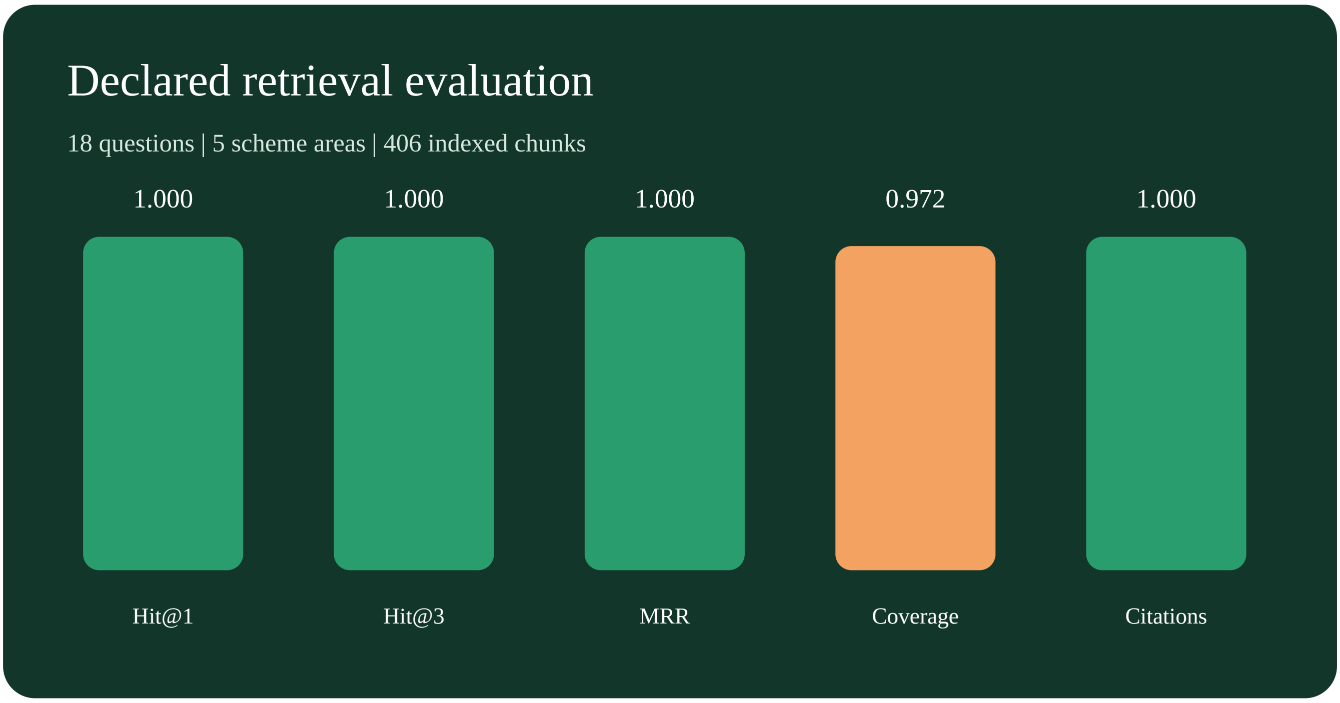
Component	Configuration	Role
Word TF-IDF	Unigrams and bigrams, 68% weight	Exact scheme and policy phrases
Character TF-IDF	3-5 character n-grams, 32% weight	Spelling and mixed-language robustness
Query expansion	Transparent Hindi and Hinglish dictionary	Domain term bridging
Sentence ranker	Question-to-sentence TF-IDF	Select concise evidence
Answer contract	At most two statements plus citations	Traceable output

Why extractive

For benefit amounts, premiums, eligibility, and credit guidance, a fluent unsupported answer is more harmful than a short official excerpt. The current design keeps every factual answer attached to inherited source metadata.

Evaluation design and results

The declared set contains 18 source-labelled questions across five scheme areas, including two Hindi questions and one Hinglish question. Each item has an expected source and evidence terms.



The expected official source ranks first for every declared question. Expected-term coverage is 0.972 because PDF extraction and chunk boundaries can separate a phrase even when the correct page is retrieved. The test is curated and does not establish general question-answering accuracy.

Citation-grounded answer example

Question: How often does a farmer receive a Soil Health Card?

It will be made available once in a cycle of 3 years, which will indicate the status of soil health of a farmer's holding for that particular period. [1] The SHC given in the next cycle of 3 years will be able to record the changes in the soil health for that subsequent period. [1]

Citation field	Example
Source	Soil Health Card FAQ: issue cycle
Publisher	Department of Agriculture and Farmers Welfare, Government of India
URL	https://support.soilhealth.dac.gov.in/kb/faq.php?id=40

The assistant returns a supported passage, source metadata, and a freshness warning. It does not request Aadhaar, banking details, land records, or personal eligibility profiles.

Limitations and responsible use

Limitation	Effect	Improvement
Small source set	Cannot answer every scheme question	Add governed sources
Curated test	Metrics may overstate general performance	Independent evaluation
Lexical retrieval	May miss semantic paraphrases	Compare embeddings
No confidence threshold	Weak matches may still return	Add abstention
Policy freshness	Rules can change	Version monitoring
PDF tables	Extracted rows can be dense	Structured table parsing

Responsible-use boundary

- Do not treat an answer as an eligibility or approval decision.
- Do not use the assistant as legal, financial, or insurance advice.
- Always verify current amounts, dates, and application steps at the cited source.
- Do not claim broad multilingual understanding from three cross-script test items.

Reproducibility and evidence

Step	Command or artifact
Environment	uv sync
Build corpus	uv run python scripts/build_knowledge_base.py
Evaluate	uv run python scripts/evaluate_retrieval.py
Create examples	uv run python scripts/demo_queries.py
Verify	uv run pytest -q and uv run ruff check .
Evidence	reports/retrieval_evaluation.json and sample_answers.json

Skills demonstrated

- Official-source acquisition, checksums, PDF and HTML extraction, and chunking.
- NLP vectorisation, hybrid ranking, mixed-language query handling, and citations.
- Retrieval evaluation, error analysis, tests, packaging, and safety documentation.

Primary official publishers

- Ministry of Agriculture and Farmers Welfare and programme portals.
- Press Information Bureau, Government of India.
- Reserve Bank of India.